ISyE 6740 – Summer 2026
Final Report

Team Member Names: Shreya Kola, Rahi Patel, Kunj Shah (Group 010)

Project Title: Are Prediction Markets Well Calibrated at Open? A Category-Level Study of First Traded Prices on Kalshi

1 Problem Statement

A prediction market assigns a price between 0 and 1 to an uncertain binary event. That price is often interpreted as the market's estimate of the probability that the event will occur. A market is calibrated when its prices match observed outcomes over many contracts. For example, events priced near 0.70 should occur about 70% of the time. Calibration is evaluated across a group of forecasts rather than from one contract alone [1, 2]. Most earlier prediction-market studies focus on prices near expiration, when the outcome is already becoming clear [4]. We instead focus on prices near the beginning of trading, when less information has entered the market.

We study Kalshi, a CFTC-regulated exchange for event contracts. Our analysis addresses three questions. First, how well calibrated are the earliest traded prices? Second, can the remaining errors be explained by observable market characteristics? Third, if prices move in a predictable direction after the first trade, would a trading strategy remain profitable after fees and spreads? Before running the analysis, we expected early prices to be useful but not perfectly calibrated. We also expected most apparent trading profits to shrink after transaction costs were included.

The final sample contains 63,498 resolved markets from November 2024 through June 2026. Most observations come from Politics, Sports, Crypto, Economics, Financials, and Mentions, with smaller groups from Science & Technology and Entertainment. Overall, the first traded prices perform well: the expected calibration error is 3.3%. However, the errors are not evenly distributed. Crowd-driven categories show an overconfidence pattern, while Crypto and other market-made categories are closer to calibration. A strategy based on these patterns earns 4.4 cents per contract before accounting for the opening spread, but loses 30 cents per contract under the conservative spread assumption.

2 Related Work

Brier [1] introduced the Brier score for evaluating probability forecasts, and Murphy [2] later separated that score into reliability, resolution, and uncertainty. We use the Brier score and expected calibration error to measure how closely prices match outcomes [3]. Previous research generally finds that prediction-market prices are informative and fairly well calibrated near expiration [4, 5]. Betting-market research also documents the favorite–longshot bias, in which low-probability outcomes tend to be overpriced and high-probability outcomes underpriced [6, 7]. Our project differs by examining the first traded price on one regulated exchange, comparing several market categories, and testing whether measured pricing errors could be traded after costs.

3 Data Collection and Cleaning

3.1 Source and ingestion

We collected the data from Kalshi's public REST API. Kalshi stores recent and older markets in separate live and historical endpoints, so the collection code checks the market date and uses the appropriate endpoint. The pipeline then combines the two sources and removes duplicate tickers. For each resolved market, we kept the outcome, opening and closing times, volume, open interest, series ticker, and category. We also collected minute- or hour-level candlesticks containing prices, volume, open interest, and bid–ask quotes.

3.2 Filters and sample construction

Table 1 summarizes the cleaning process. The API walk scanned about 56 million settled-market records. We removed multivariate parlay wrappers because they combine other markets and would duplicate outcomes. We also removed nonbinary contracts, voided markets, observations outside the November 2024–June 2026 window, and categories outside the retained scope. After these steps, 9,956,482 binary markets remained.

Many of the remaining contracts were ladder strikes that never traded. In a preliminary sample, about 96% of Crypto and Financials contracts had no trades. Pulling detailed candlesticks for every inactive contract would add substantial collection time without improving the calibration analysis. We therefore selected a seeded, category-stratified sample of up to 20,000 markets per category among contracts with at least 100 units of lifetime volume. This produced 96,884 markets, and 96,461 had a first traded price. For the main analysis, we further required the first trade to occur within the first 25% of the market's lifetime, leaving **63,498 observations**.

Step	Markets	Note
Settled-market records scanned	~56,000,000	live + historical tiers
excl. multivariate parlay wrappers	−46,322,126	combos of other markets
excl. no yes/no result	−22,821	voided / non-binary results
excl. category out of scope	−239,066	series outside retained scope
Unit binary markets in scope	9,956,482	Nov 2024 – Jun 2026
Stratified subsample (traded, $\leq$20k/cat.)	96,884	seeded, reproducible
with a first traded price	96,461	99.6%
Core sample (early first trade)	**63,498**	first trade $\leq$ 25% of lifetime

Table 1: Sample construction and cleaning. Each exclusion is logged by the pipeline; the stratified subsample is drawn with a fixed seed and is reproducible.

3.3 A census finding that reshaped the design

Our proposal planned to use an average price from a fixed early window, such as the first hour. During data collection, we found that this definition did not work well for Kalshi. Outside Crypto and Financials, opening spreads were often between 0.62 and 0.98, and fewer than one-quarter of markets traded during their first hour. Sports markets were especially late, with a median first trade about 10 hours after opening. A first-hour average would therefore remove most of the sample and would often describe quotes rather than completed trades.

Based on that finding, we changed the primary measure to the first traded price. This is the earliest price backed by an actual transaction. We also kept two alternatives for comparison: the first-day VWAP and the VWAP from the first 10% of a market's lifetime. The second measure helps compare short Crypto contracts with political markets that may remain open for months. As shown later, the choice of window changes the measured bias, so this adjustment was an important part of the analysis rather than only a data-cleaning decision.

4 Methodology

Throughout, $p_i \in (0, 1)$ denotes market i's early price under a given definition and $y_i \in \{0, 1\}$ its settlement outcome.

4.1 Calibration (Workstream 1)

We group markets into ten probability bins by early price and compare each bin's empirical resolution rate against its mean price (a reliability diagram). Aggregate accuracy is the Brier score $\mathrm{BS} = \frac{1}{n}\sum_i (p_i - y_i)^2$, which we compare to the no-skill baseline $\bar{y}(1-\bar{y})$ (always forecasting the base rate) and decompose per Murphy [2] as BS = reliability − resolution + uncertainty, attributing error to miscalibration vs. lack of discrimination. Binned miscalibration is summarized by the expected calibration error $\mathrm{ECE} = \sum_b \frac{n_b}{n} |\bar{y}_b - \bar{p}_b|$ [3].

Uncertainty quantification. Contracts from the same event are not independent. For example, several election strikes or the two sides of a game may share the same underlying outcome. Resampling individual contracts would make the confidence intervals too narrow. We therefore used a cluster bootstrap that resamples complete events. We used 2,000 replications for the main statistics and 1,000 for the bin-level confidence intervals.

4.2 Favorite–longshot bias (Workstream 2)

We test the classical favorite–longshot pattern [6] two ways. First, a nonparametric statistic $\Delta = \overline{(y-p)}_{p\geq 0.8} - \overline{(y-p)}_{p\leq 0.2}$ (mean deviation among favorites minus longshots), positive under the classic bias, with a cluster-bootstrap two-sided p-value. Second, a parametric recalibration fit $y \sim \sigma(a + b\,\mathrm{logit}(p))$ estimated by Newton–Raphson: $(a, b) = (0, 1)$ under perfect calibration, $b < 1$ indicating overconfident prices (outcomes less extreme than prices imply), $b > 1$ underconfident.

4.3 Residual predictability (Workstream 3)

Under efficiency, y_i is unpredictable given p_i beyond p_i itself. Our baseline is deliberately stronger than a naive base-rate model (sharpening a peer-review request to define the baseline precisely): a price-only logistic recalibration $y \sim \sigma(a + b\,\mathrm{logit}(p))$, which already corrects any aggregate miscalibration. Against it we fit three models that combine the price with market features (category, early volume, open interest, time to resolution, opening spread, first-trade timing, and open-time features): ℓ_2-regularized logistic regression, a random forest [8], and gradient-boosted trees [9] with early stopping. Evaluation uses expanding-window temporal cross-validation (5 folds ordered by close time): a model is always evaluated on markets that resolve strictly after all of its training data, excluding temporal leakage. We report held-out AUC and log loss, gains over the price-only baseline, and permutation feature importances.

4.4 Drift and tradeability (Workstream 4)

A market's price at resolution equals its settlement value, so open-to-resolution drift is the residual $y_i - p_i$. We estimate mean drift per (category $\times$ price bin) cell with cluster-bootstrap intervals, then assess exploitability with a walk-forward backtest: trading rules are learned on the earliest 70% of markets (cells whose drift interval excludes zero, minimum 50 training markets) and evaluated strictly out-of-sample on the remaining 30%. Execution realism follows Kalshi's actual cost structure: the per-contract trading fee $0.07\,\pi(1-\pi)$ at entry price π [10]. Because the true spread at first-trade time is unobserved, we use two bounding cost scenarios: fees only (lower bound), and fees plus half the opening quoted spread (a conservative upper bound, since books tighten between open and first trade). We report gross and net edge per contract with cluster-bootstrap intervals under both scenarios.

5 Evaluation and Final Results

5.1 Calibration results for first traded prices

Table 2 presents the main calibration results. The overall Brier score is 0.175, compared with 0.246 for a model that always predicts the sample base rate. The Murphy decomposition gives a reliability term of 0.0017 and a resolution term of 0.072. The expected calibration error is 3.3%. Figure 1 shows the same result across the ten price bins, along with event-cluster bootstrap confidence intervals.

The results also differ by category. Crypto has an ECE of 0.8%, and Financials has an ECE of 1.6%, so both are close to calibration. Mentions has the largest ECE among the major categories at 9.0%. Politics, Economics, and Sports fall between these groups at 3.8%, 3.9%, and 3.1%, respectively. Science & Technology and Entertainment have errors above 10%, but these estimates are based on much smaller samples.

Scope	n	Brier	No-skill	ECE	Reliability	Resolution
All markets	63,498	0.175	0.246	0.033	0.0017	0.072
Crypto	16,386	0.152	0.244	0.008	0.0001	0.091
Mentions	14,906	0.214	0.249	0.090	0.0095	0.045
Financials	11,759	0.175	0.250	0.016	0.0003	0.074
Sports	7,115	0.187	0.234	0.031	0.0027	0.050
Economics	6,256	0.151	0.250	0.039	0.0016	0.100
Politics	6,153	0.158	0.216	0.038	0.0035	0.062
Science & Tech.	732	0.146	0.250	0.107	0.0214	0.126
Entertainment	169	0.127	0.171	0.124	0.0320	0.075

Table 2: Calibration of first traded prices. "No-skill" is the base-rate Brier score $\bar{y}(1-\bar{y})$. Reliability and resolution are the Murphy decomposition terms (lower reliability and higher resolution are better).

5.2 Reverse favorite–longshot pattern by category

Table 3 and Figure 2 show a pattern opposite to the usual favorite–longshot bias. For the full sample, the difference between favorite and longshot deviations is -0.081, with a 95% confidence interval from -0.090 to -0.072 and $p < 0.001$. The logistic recalibration slope is 0.80, with a confidence interval from 0.79 to 0.82. These values indicate that the first traded prices are too extreme: favorites are priced too high and longshots too low.

This pattern is strongest in the categories with thinner, more crowd-driven markets. The favorite–longshot deviation is -0.169 for Sports, -0.173 for Mentions, and -0.135 for Politics. Crypto is different. Its deviation is -0.006 with $p = 0.33$, and its recalibration slope is 0.98 with a confidence interval that includes 1. This category comparison suggests that professional liquidity provision may help correct extreme first prices.

The result is sensitive to how the early price is defined. The slope increases from 0.80 for the first trade to 1.72 for the first-day VWAP and 1.35 for the first-10%-of-life VWAP. For short contracts, a full day can include most or all of the market's life, so the VWAP contains later prices that are already moving toward the outcome. This is especially clear for Crypto. The first-trade measure is therefore the best fit for our question about initial pricing, while the wider windows help show how price discovery develops.

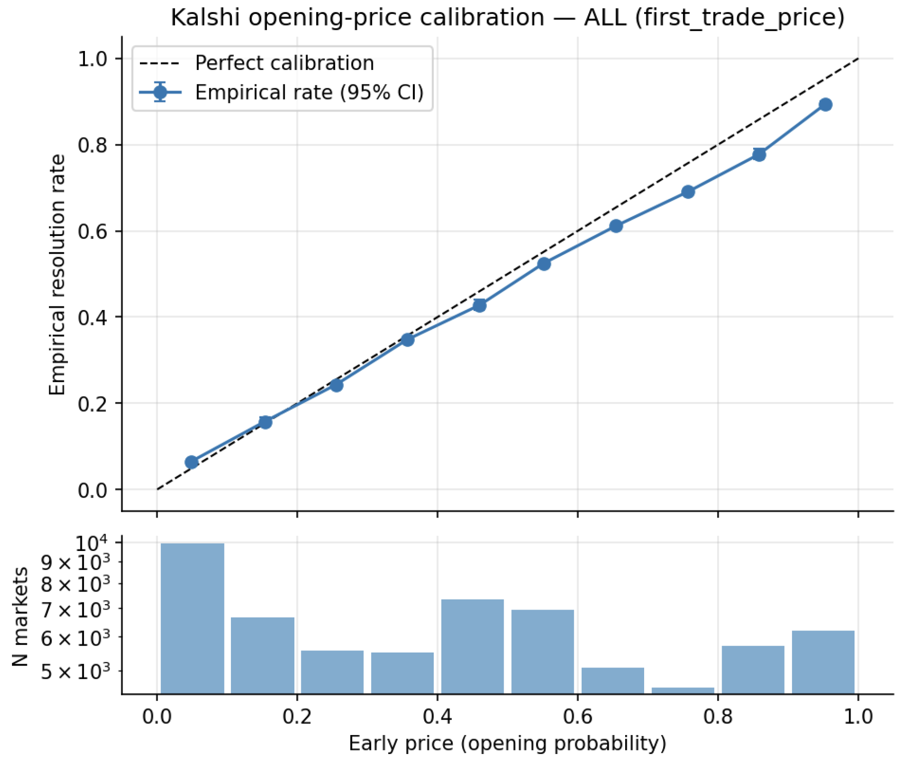

Figure 1: Reliability diagram for first traded prices, all categories pooled ($n = 63{,}498$). Whiskers are 95% event-cluster bootstrap intervals on bin-level resolution rates; the lower panel shows the market count per bin (log scale).

5.3 Limited improvement from additional market features

Table 4 compares the models using expanding-window temporal cross-validation. The price-only logistic model has a mean AUC of 0.795. Random forest reaches 0.799, while gradient boosting also reaches 0.799 and has the lowest log loss at 0.537. The improvement over the price-only model is consistent but small. In practice, the first traded price already contains most of the available predictive information. Permutation importance shows that early volume, opening spread, category, and time to resolution explain some of the remaining error.

5.4 Tradeability after fees and the opening spread

The walk-forward backtest identified 24 category-by-price-bin cells with significant drift during the training period. Applying those rules to the later test period produced 7,865 simulated trades. Table 5 reports the average result under the two cost assumptions. Because the traded signal is the same deviation pattern measured in Section 5.2, this backtest also serves as a strict out-of-sample validation of the reverse favorite–longshot finding: the pattern learned on the earliest 70% of markets persists in the later 30%.

Under the fees-only assumption, the strategy earns 0.044 dollars per contract, with a 95% confidence interval from 0.032 to 0.055. Politics, Mentions, Economics, and Financials have positive results, while Crypto and Sports are not significantly different from zero. This shows that the pricing pattern has economic size before spreads are included. The result changes under the conservative assumption that a trader pays half of the opening spread. Average net profit then falls to -0.300

Scope	n	Δ (fav. − longshot dev.)	95% CI	Logit slope b
All markets	63,498	−0.081	[−0.090, −0.072]	0.80
Crypto	16,386	−0.006	[−0.019, +0.007]	0.98
Mentions	14,906	−0.173	[−0.198, −0.148]	0.68
Financials	11,759	−0.037	[−0.059, −0.017]	0.88
Sports	7,115	−0.169	[−0.208, −0.131]	0.75
Economics	6,256	−0.082	[−0.110, −0.054]	0.76
Politics	6,153	−0.135	[−0.183, −0.093]	0.74

Table 3: Favorite–longshot analysis of first traded prices. Negative Δ and slope $b < 1$ indicate reverse favorite–longshot bias (overconfident prices). Categories with $n < 1{,}000$ omitted; all $p < 0.001$ except Crypto ($p = 0.33$).

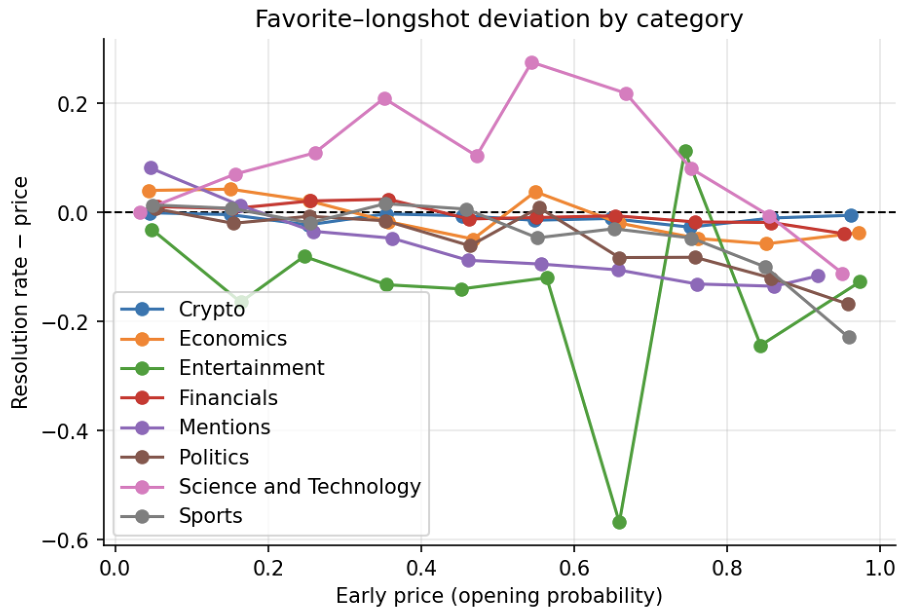

Figure 2: Signed deviation (resolution rate minus price) by price bin and category, first traded prices. The downward tilt, positive at low prices and negative at high prices, is the signature of reverse favorite–longshot bias.

dollars per contract, and no category remains significantly profitable.

The two scenarios lead to a practical conclusion. Some first prices appear to be mispriced, but the same markets usually have wide opening spreads. A trader may be able to identify the direction of the error without being able to trade it profitably. Because the exact spread at the moment of the first trade is not available, the true result lies somewhere between the fees-only and opening-spread estimates.

6 Limitations, Threats to Validity, and Future Work

Selection. We can only measure calibration for markets that actually produced a price. The main sample also requires at least 100 contracts of lifetime volume and a first trade within 25% of the market's lifetime. Our conclusions therefore apply to actively traded Kalshi markets, not to the many ladder strikes that never traded. We did not run a separate sensitivity analysis for the 25%

Model	AUC (mean ± sd)	Log loss	ΔAUC vs. baseline	ΔLL vs. baseline
Price-only logistic (baseline)	0.795 ± 0.007	0.543	—	—
ℓ_2 logistic + features	0.796 ± 0.007	0.542	+0.0005	+0.0010
Random forest	0.799 ± 0.006	0.543	+0.0040	+0.0003
Gradient boosting	0.799 ± 0.006	0.537	+0.0032	+0.0063

Table 4: Residual predictability under 5-fold expanding-window temporal cross-validation (n = 63,498). Top permutation importances (beyond the price itself): early volume, opening spread, category, time to resolution.

Cost scenario	Scope	n trades	Gross edge	Net edge	95% CI
Fees only (lower bd.)	All	7,865	+0.054	+0.044	[+0.032, +0.055]
	Politics	178	+0.141	+0.135	[+0.051, +0.220]
	Mentions	3,747	+0.082	+0.069	[+0.051, +0.087]
	Economics	1,914	+0.036	+0.031	[+0.012, +0.052]
	Financials	244	+0.047	+0.044	[+0.002, +0.091]
	Sports	1,499	+0.010	−0.002	[−0.023, +0.020]
	Crypto	282	+0.003	−0.010	[−0.061, +0.040]
Fees + open spread (upper bd.)	All	7,865	−0.290	−0.300	[−0.315, −0.287]
	Politics	178	+0.015	+0.005	[−0.095, +0.105]
	Financials	244	−0.023	−0.029	[−0.081, +0.022]

Table 5: Out-of-sample backtest of the drift-following rule under bounding transaction-cost scenarios (per-contract dollar edge; 95% event-cluster bootstrap intervals on the net edge). Upper-bound rows for the remaining categories are all strongly negative and omitted for space; full tables in the repository.

cutoff, although the alternative price windows provide some evidence about timing.

Cost measurement. We do not observe the bid–ask spread at the exact time of the first trade. For this reason, we report a lower-cost case with fees only and a conservative case using half of the opening spread. Actual costs should fall between these values. The backtest also assumes that trades can be filled at the quoted price without market impact, which may be optimistic for thin markets.

Category coverage. We use the category labels supplied by Kalshi. Some election markets are listed separately under Elections rather than Politics, so the Politics sample does not cover every election contract. The historical endpoints also limited the sample to markets available from November 2024 onward.

Price definition. The direction of the measured bias changes when the price window changes. We use the first traded price because it most directly represents the market's initial transaction, but we report the two VWAP alternatives so that this choice is visible rather than hidden.

Measurement noise. A single first trade is a noisy measurement of the market's belief, and noise in the price variable mechanically attenuates the recalibration slope below 1 (a standard errors-in-variables effect). Part of the estimated overconfidence could therefore reflect price noise rather than belief bias. Two observations suggest the effect is not purely mechanical: Crypto, whose market-made first prices should be the least noisy, shows a slope of 0.98, and pure symmetric noise would not generate the significantly positive out-of-sample fee-only trading edge in Table 5. Separating noise from bias more formally, for example by instrumenting the first trade with the second, is left to future work.

Future work. A useful next step would be to follow the full price path from the first trade to settlement and estimate when the early overconfidence disappears. Collecting order-book depth and the spread at the exact first-trade time would also narrow the transaction-cost range. Finally, trader-level or order-level data could help explain why the first quote is more extreme in some categories than in others.

7 Conclusion

We analyzed 63,498 resolved Kalshi markets to evaluate how much information is contained in the first traded price. The overall Brier score is 0.175, compared with 0.246 for the no-skill baseline, and the expected calibration error is 3.3%. The remaining error follows a reverse favorite–longshot pattern that is strongest in thin, crowd-driven categories and nearly absent in Crypto. Adding market characteristics to the price produces only a small improvement in out-of-sample prediction. The backtest earns 4.4 cents per contract in the fees-only case but loses 30 cents when the conservative opening-spread cost is applied. Our main finding is therefore that early prices are not perfectly calibrated, but the observed errors are difficult to trade after realistic costs.

8 Team Responsibilities

Shreya Kola: data extraction and cleaning, including the Kalshi API client, live/historical ingestion, census, and subsample construction. **Rahi Patel**: calibration and favorite–longshot analysis, including reliability diagrams, Brier/ECE and decomposition, bias tests, and figures. **Kunj Shah**: residual-prediction modeling and drift/tradeability, including temporal cross-validation, model comparison, backtest, and cost scenarios. All members contributed to methodology decisions (including the census-driven early-price redesign), interpretation, and writing. Code: `https://github.com/Kunj892/kalshi-calibration`.

References

[1] Brier, G. W. (1950). Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1), 1–3.

[2] Murphy, A. H. (1973). A new vector partition of the probability score. *Journal of Applied Meteorology*, 12(4), 595–600.

[3] Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017). On calibration of modern neural networks. *Proceedings of ICML.*

[4] Wolfers, J., & Zitzewitz, E. (2004). Prediction markets. *Journal of Economic Perspectives*, 18(2), 107–126.

[5] Page, L., & Clemen, R. T. (2013). Do prediction markets produce well-calibrated probability forecasts? *The Economic Journal*, 123(568), 491–513.

[6] Thaler, R. H., & Ziemba, W. T. (1988). Anomalies: Parimutuel betting markets. *Journal of Economic Perspectives*, 2(2), 161–174.

[7] Snowberg, E., & Wolfers, J. (2010). Explaining the favorite–longshot bias: Is it risk-love or misperceptions? *Journal of Political Economy*, 118(4), 723–746.

[8] Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.

[9] Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5), 1189–1232.

[10] Kalshi, Inc. (2026). Fee schedule and API documentation. `https://docs.kalshi.com`.